

FOGNET 4.1

Cooperative Safety and Haulage Continuity Network for Open-Cast Mines

Smart India Hackathon	2026
Problem Statement ID	26007
Theme	Smart Automation
Category	Hardware
Target Organization	Ministry of Steel / NMDC Limited
Application Context	Bailadila Iron Ore Complex, Chhattisgarh

SEE → SHARE → CALCULATE → WARN → LOG

A complete solution, architecture, prototype plan, demonstration plan and judge-defense document.

1. THE ONE-LINE PITCH

FogNet 4.1 is a low-cost cooperative mine-safety network that combines vehicle-mounted radar, strategic roadside sensing, worker safety tags, vehicle-to-vehicle communication, and physics-based dynamic safe-speed calculation to help haul trucks operate safely and continuously during dense fog and low-visibility conditions.

AI predicts. Physics protects. Infrastructure sees beyond the driver. Local safety never depends on the network.

2. THE PROBLEM

Open-cast iron ore mines depend heavily on haul trucks and dumpers to continuously move ore between excavation faces, crushers, stockyards, and processing facilities.

At Bailadila, monsoon conditions can produce heavy rain, humidity, cloud and dense fog. In severe conditions, visibility can fall to only a few metres.

The driver can lose reliable visual awareness of approaching vehicles, blind bends, road obstructions, workers and spotters, temporary hazards, road edges and dangerous sections.

This creates a safety problem: drivers may not have enough information to react safely to a hazard hidden by fog.

It also creates a productivity problem: drivers may respond by drastically reducing speed or stopping the haulage operation.

The mine needs a way to increase situational awareness without requiring every truck to become an expensive autonomous vehicle.

FogNet creates a cooperative safety layer around the existing haul-road operation.

3. WHAT FOGNET ACTUALLY DOES

FogNet is not designed to replace the truck driver. It provides the driver and mine supervisor with better information about what cannot be seen reliably in fog.

1. SEE

Vehicle and roadside sensors detect vehicles, obstacles, workers and environmental conditions.

2. SHARE

Vehicles, workers and roadside nodes exchange compact safety information.

3. CALCULATE

The vehicle estimates whether available clearance is sufficient for current speed and road conditions.

4. WARN

The driver receives simple visual, acoustic and haptic alerts.

5. LOG

Important events are recorded for later analysis and mine-road improvement.

SEE → SHARE → CALCULATE → WARN → LOG

4. THE KEY INNOVATION

The main innovation is not radar alone, V2V alone, or AI alone. Those technologies already exist individually.

The innovation is their combination into a cost-optimized cooperative architecture.

Every truck receives a Safety Core for local obstacle awareness, vehicle state, safety calculation, direct communication and driver alerts.

Strategic road locations receive Road Guardians for visibility monitoring, environmental sensing, additional perception, traffic coordination and local hazard information.

Workers receive Safety Tags for active identification and warning where visual detection is unreliable.

The control room receives fleet monitoring, risk maps, fog corridors, near-miss analysis and productivity information.

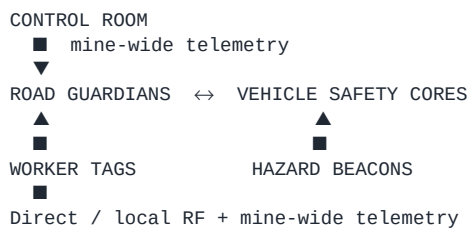
Make the truck affordable. Make the road intelligent.

5. SYSTEM ARCHITECTURE

FogNet consists of four cooperative layers: Tier 1 Vehicle Safety Core, Tier 2 Road Guardians, Tier 3 Worker Safety / Temporary Hazard Beacons, and the Control Room.

The control room uses a lightweight haul-road digital twin, risk map, fog corridors, near-miss analysis and fleet/production monitoring.

Road Guardians exchange local V2X information with vehicles. Worker tags and hazard beacons provide additional field-level safety information.



6. TIER 1 — VEHICLE SAFETY CORE

Every participating haul truck receives a compact local safety unit. The prototype demonstrates the architecture with student-available components; industrial deployment uses ruggedized, appropriate hardware.

Radar

Industrial deployment: 76–81 GHz FMCW radar for obstacle range, relative velocity, approaching-vehicle detection and operation during poor optical visibility.

The prototype demonstrates: sensor → processor → safety decision → driver alert.

A low-cost development radar should not be claimed to have the same performance as an industrial 77 GHz mining radar.

IMU

Provides acceleration, angular velocity, vehicle pitch, roll and short-duration motion information. Pitch matters because stopping behaviour changes with road gradient.

GNSS

Provides position, ground-speed estimate, road-segment identification, event location and fleet tracking. IMU can support short-duration dead reckoning where GNSS is unreliable.

Edge Processor

An ESP32-class controller can be used for the prototype. The local processor handles sensor acquisition, filtering, safety calculations, communication, state-machine logic, driver alerts and event logging.

7. TIER 2 — ROAD GUARDIANS

A Road Guardian is a fixed intelligent roadside node installed only at strategically selected high-risk locations: blind bends, steep ramps, intersections, crusher approaches, narrow road sections, high-risk pedestrian areas and recurring fog locations.

Environmental sensing can include visibility/light attenuation, humidity and temperature. Perception can include thermal camera, conventional camera, radar or other fixed sensing depending on site requirements.

The node communicates fog conditions, hazards, traffic information, road-state information and warnings.

At a narrow junction it can operate a logical traffic-control function using direction, position, speed, road segment and site-defined priority rules.

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TRUCK A → approaching junction
TRUCK B → opposite direction
      ↓
Road Guardian evaluates:
Direction / Position / Speed / Road segment / Priority
      ↓
A: HOLD      B: PROCEED

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8. TIER 3 — ACTIVE WORKER SAFETY

Workers can carry small active safety tags that periodically broadcast an identifier.

A nearby truck can detect the tag and determine whether the worker is entering a dangerous area.

The worker warning zone should not be a universal fixed distance.

The warning envelope can depend on truck speed, estimated stopping distance, direction of movement, road geometry, visibility, worker position and predicted trajectory.

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Truck speed increases
      ↓
Stopping distance increases
      ↓
Worker safety envelope increases

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The prototype can implement a simplified version; industrial deployment requires accurate positioning and validated radio/perception performance.

9. TIER 3 — TEMPORARY HAZARD BEACONS

Mining conditions can change rapidly: stalled machines, maintenance vehicles, washed-out shoulders, blast exclusion zones, damaged road sections or temporary obstructions can appear.

A supervisor can deploy a temporary hazard beacon so nearby vehicles receive a warning without waiting for a central map update.

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Temporary hazard → supervisor activates beacon
→ nearby vehicles receive message → driver warning → event logged

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10. COMMUNICATION ARCHITECTURE

Communication is a safety enhancement, not the sole safety mechanism.

Level 1 — Local vehicle communication

Nearby truck awareness, worker alerts, Road Guardian interaction and immediate cooperative warnings. The prototype can use ESP-NOW or another suitable local radio.

Level 2 — Mine-wide telemetry

Corridor information, fleet status, road condition, fog status and event synchronization. LoRa or another suitable network can be evaluated.

Level 3 — Control room

Fleet visualization, event logging, analytics, risk mapping and production monitoring.

11. THE GOLDEN SAFETY RULE

A network failure must never remove the vehicle's basic local safety capability.

If V2V fails, local sensing continues. If a Road Guardian fails, the onboard Safety Core continues. If the control room or cloud connection fails, local safety continues.

NETWORK AVAILABLE → more information / better awareness / better coordination
 NETWORK LOST → local safety remains active

12. THE DYNAMIC SAFETY BUBBLE

This is the central technical concept of FogNet. Instead of one fixed speed limit for every situation, FogNet calculates whether the truck has enough available distance to respond safely.

$$D_{\text{required}} = v \cdot t_{\text{reaction}} + \frac{v^2}{2a_{\text{safe}}} + D_{\text{buffer}}$$

Where v is current speed, t_{reaction} is assumed driver/system response time, a_{safe} is conservative validated deceleration, and D_{buffer} is additional safety margin.

Available distance comes from onboard perception, cooperative vehicle information, known road hazards and roadside infrastructure.

$$\text{Safety Margin} = D_{\text{available}} - D_{\text{required}}$$

13. WHY THIS IS BETTER THAN A FIXED SPEED LIMIT

Two trucks may experience very different conditions. Truck A may be slow on a level, good surface with large clearance. Truck B may be faster, downhill, wet, visibility-limited and approaching another vehicle.

Given the current situation, how much stopping distance do I actually have?

FogNet uses the current situation to determine an appropriate operating state rather than applying one generic speed recommendation.

14. SAFETY STATES

■ NORMAL

Conditions are acceptable; normal monitoring.

■ ADVISORY

Conditions are deteriorating: decreasing visibility, increasing moisture, approaching fog corridor or increasing traffic density. Recommend lower safe approach speed.

■ RESTRICTED

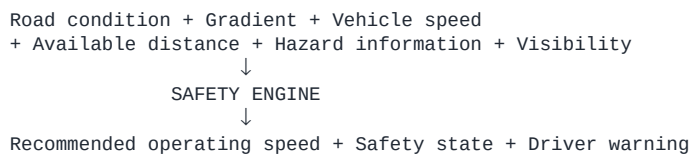
Elevated road risk; possible lower speed, traffic metering, one-way sequencing, controlled entry and supervisor notification.

■ EMERGENCY

Safety margin is critically low or an immediate hazard is detected. Provide high-priority alert and request immediate driver action.

Automated vehicle-control functions beyond prototype demonstration require OEM integration, validation and regulatory approval.

15. SAFE-SPEED ENGINE



The production system does not assume one universal friction coefficient. Road friction varies with surface material, moisture, mud, slope, tyre condition, loading and maintenance. Production parameters should therefore be site-calibrated and conservative.

16. EXPLAINABLE DRIVER ALERTS

The driver should understand why the system is warning them.

CRITICAL HAZARD
 Hazard: Oncoming vehicle
 Available clearance: 42 m
 Required stopping distance: 49 m
 Current state: EMERGENCY
 Recommended action: DECELERATE / STOP

Explainable warnings improve trust and make the system easier to evaluate.

17. DRIVER HMI

Green

SAFE / NORMAL

Yellow

CAUTION — Recommended speed: 15 km/h

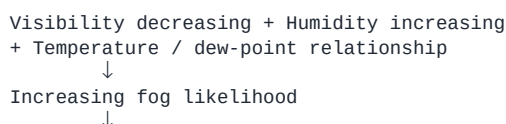
Red

CRITICAL — DECELERATE / STOP

Additional feedback can include buzzer, LEDs, vibration motor and a small OLED. Industrial directional haptic feedback can indicate left, centre or right hazard.

18. PREDICTIVE FOG CORRIDORS

FogNet does not need to claim exact fog prediction. Road Guardians monitor changing environmental conditions.



Road segment marked ADVISORY



Approaching trucks receive warning

FOG CORRIDOR ACTIVE

Segment: Ramp 4

Observed visibility: 8 m

Recommended approach speed: 14 km/h

State: ADVISORY

Exact thresholds are configurable and must be calibrated for the deployment site.

19. COOPERATIVE VEHICLE AWARENESS

Participating trucks can periodically broadcast compact safety messages containing truck ID, position/road segment, speed, direction, load state, safety state and timestamp.

A truck can receive information from another truck even when it is temporarily hidden behind a bend. Cooperative information is additional evidence; local perception remains the primary immediate safety mechanism.

20. TRAFFIC PRIORITY

FogNet can support site-defined traffic rules at narrow sections.

TRUCK A – PRIORITY / PROCEED

TRUCK B – HOLD / YIELD

Traffic priority is a configurable operational rule, not a universal claim that every downhill loaded truck must always receive priority. The mine's operating and safety procedures determine the rule.

21. GRACEFUL SENSOR DEGRADATION

If a visibility sensor is unavailable, the vehicle continues with onboard sensing. If a Road Guardian is unavailable, local vehicle safety continues. If cooperative radio is unavailable, the vehicle enters a conservative local mode.

If radar is unavailable, the prototype can demonstrate degraded/crawl mode and request operator/supervisor intervention. If multiple critical sensors are unavailable, the system requests a controlled halt according to the defined safety policy.

The exact fallback sequence must be validated for the actual vehicle and sensor set.

22. THE FOUR-LAYER SAFETY PHILOSOPHY

LAYER 1 – Local vehicle perception

"Is something immediately in front of me?"

LAYER 2 – Cooperative awareness

"Is another vehicle / worker / hazard nearby?"

LAYER 3 – Road infrastructure

"What is happening beyond my local view?"

LAYER 4 – Control room

"What is happening across the whole mine?"

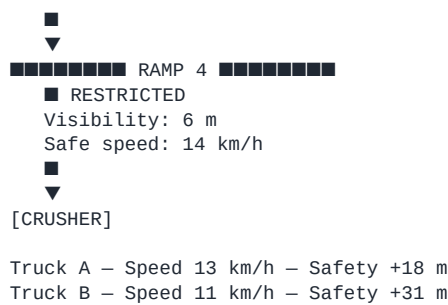
If Layer 4 disappears, Layer 1 does not disappear.

23. HAUL-ROAD DIGITAL TWIN

FogNet intentionally uses a lightweight 2D spatial representation rather than an incomplete 3D mine model.

It shows haul-road segments, trucks, Road Guardians, workers, hazards, fog corridors, safety states, risk levels and recent events.

[SHOVEL]



24. NEAR-MISS INTELLIGENCE

Important events can be logged: hard deceleration, close vehicle approach, worker warning, blind-curve conflict, communication failure, sensor failure, emergency warning and fog-corridor activation.

Each event can contain timestamp, location, vehicle, speed, hazard, safety margin, road condition, sensor state and communication state.

Over time this creates a Near-Miss Risk Heatmap, helping supervisors identify road sections that repeatedly create dangerous situations.

25. OPERATIONAL CONTINUITY

FogNet is also an operational information system. The control room can monitor average truck speed, haul cycle time, queue time, fog-related slowdowns, restricted road segments, fleet availability, safety interventions and near-miss frequency.

Without FogNet

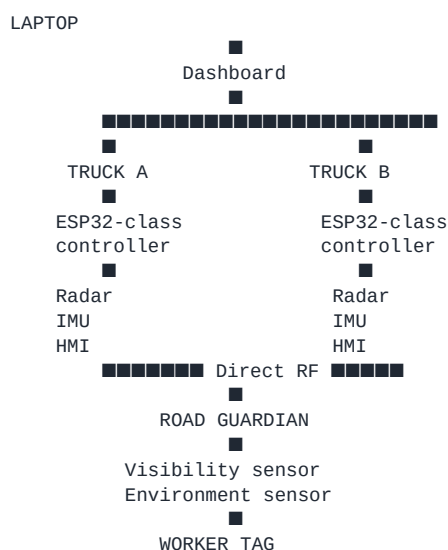
Fog → Driver uncertainty → Extreme slowdown / stoppage → Longer cycle time → Production disruption

With FogNet

Fog → Early warning → Controlled speed reduction → Coordinated traffic → Safer continuous haulage

Actual productivity improvement must be measured during field trials rather than assumed.

26. PROTOTYPE ARCHITECTURE



27. WHAT THE STUDENT PROTOTYPE MUST PROVE

The prototype does not need to prove complete mine deployment. It needs to prove the core engineering logic:

- A vehicle can detect an obstacle using its sensor.
- The system can calculate a safety condition.
- Two vehicles can exchange safety information.
- A roadside node can detect deteriorating visibility.
- A worker tag can generate a warning.
- The vehicle can continue basic safety operation when communication is lost.
- The dashboard can record and visualize safety events.

If these seven things work reliably, the prototype demonstrates the architecture convincingly.

28. PROTOTYPE HARDWARE

Vehicle A / Vehicle B

ESP32-class board, suitable radar/distance sensor, MPU6050-class IMU, RF communication, OLED, buzzer, vibration motor, battery and model vehicle.

Road Guardian

ESP32-class board, visibility/light attenuation sensor, humidity/temperature sensor, RF communication, LED and power source.

Worker Tag

ESP32-C3/nRF52-class low-power controller, RF communication, buzzer/vibration, LED and battery.

Demonstration environment

Two model vehicles, miniature road, blind bend, obstacle, transparent fog chamber, controlled mist source and laptop dashboard.

29. IMPORTANT PROTOTYPE/HARDWARE HONESTY

The prototype must clearly distinguish demonstration hardware from industrial hardware.

Demonstration hardware is low-cost equipment selected to prove the concept. Industrial hardware must be ruggedized and selected for range, vibration, dust, rain, temperature, electromagnetic compatibility, mine environment and vehicle integration.

The prototype validates the sensing and decision architecture. Industrial deployment would use a suitable certified 76–81 GHz FMCW radar.

30. VEHICLE CONTROL

The SIH prototype should not directly control the braking system of a real mining dumper.

Hazard detected → Safety engine → RED state → Model vehicle motor command reduced → Vehicle slows/stops

For actual HEMM, any throttle, retarder or vehicle-control integration requires OEM-approved interfaces, engineering validation and compliance with applicable mine-safety requirements.

FogNet therefore starts as a driver-assistance and decision-support platform; deeper vehicle integration is a future deployment phase.

31. INDUSTRIAL ROADMAP

Phase 1 — SIH Prototype

Demonstrate sensing, V2V, Road Guardian, worker tag, safety calculation, HMI, dashboard and failure handling.

Phase 2 — Controlled Mine Pilot

Pilot on one selected high-risk corridor with participating vehicles, Road Guardians, worker tags, hazard beacons and control-room dashboard. Measure detection, warning response, communication reliability, false alarms, safe-speed accuracy, sensor availability, operator acceptance, near-miss indicators and haul-cycle effects.

Phase 3 — OEM / Mine Integration

Potentially add industrial-grade radar, ruggedized electronics, OEM-approved interfaces, advanced thermal perception, mine-wide communication, improved localization, predictive analytics and dispatch integration.

32. WHAT WE ARE NOT BUILDING

To keep the SIH project feasible, FogNet deliberately does not attempt to build everything at once.

- autonomous steering
- fully autonomous dumpers
- direct hydraulic emergency braking
- expensive multi-camera systems on every truck
- LiDAR-based complete perception
- mine-wide autonomous traffic control
- a full 3D digital twin
- cloud-dependent safety logic
- a large AI model
- production-certified HEMM electronics

We are building the smallest system that proves the largest idea.

33. AI / ML ROLE

AI is deliberately limited to tasks where prediction is useful: fog trend prediction, road-segment risk scoring, near-miss pattern analysis, anomaly detection, recurring dangerous locations and operational analytics.

AI does not make the immediate safety decision.

AI → "Risk is increasing"

Physics engine → "Available distance is insufficient"

Safety state → RED

This separation improves explainability and reduces dependence on an opaque model for a critical decision.

34. WHY NOT ONLY USE AI CAMERA VISION?

Optical perception becomes more difficult as visibility deteriorates. FogNet therefore uses sensor complementarity rather than one magic sensor.

Camera / thermal → useful where visibility permits

Radar → ranging and relative-motion information
 IMU → vehicle motion
 GNSS → position
 V2V → information beyond local line of sight
 Road Guardian → infrastructure-level awareness

35. WHY NOT PUT THERMAL CAMERAS ON EVERY TRUCK?

The objective is maximum safety improvement per unit deployment cost and maintenance effort, not maximum sensor capability.

Many trucks → Low-cost Safety Core

Few dangerous locations → Higher-capability Road Guardians

Strategic Road Guardians place more expensive sensing where it can provide the greatest benefit.

36. EXPECTED BENEFITS

1. SAFETY

Better awareness of vehicles, workers, obstacles and dangerous road conditions.

2. CONTINUITY

Reduce unnecessary full-fleet stoppage during low visibility.

3. EFFICIENCY

Support controlled operating speeds and coordinated traffic.

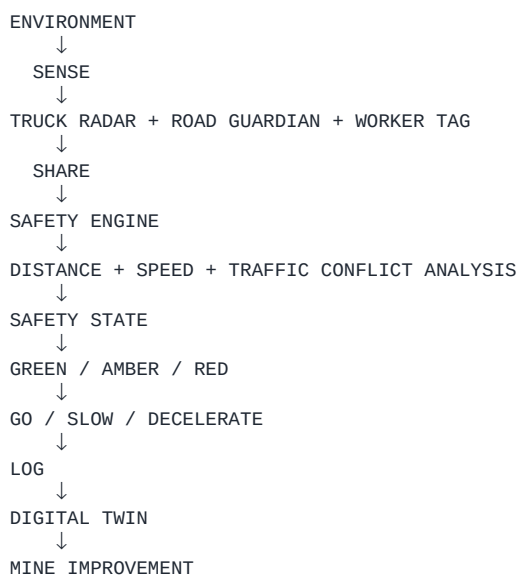
4. MONITORING

Give supervisors real-time visibility into road and fleet conditions.

5. LEARNING

Use logged near-misses and risk patterns to improve haul-road design and operating procedures.

37. THE FINAL FOGNET LOOP



38. THE 6-MINUTE SIH DEMONSTRATION

Scene 1 — Normal operation

Two model trucks move on the miniature haul road. Dashboard: NORMAL.

Scene 2 — Fog appears

Mist is introduced. Road Guardian detects reduced optical transmission. Dashboard: FOG CORRIDOR — ADVISORY. Truck display: RECOMMENDED SPEED: 15 km/h.

Scene 3 — Driver visibility fails

Camera demonstration becomes visually obscured; human optical visibility has degraded.

Scene 4 — Local sensing remains available

Truck sensor detects an obstacle. Safety engine compares available and required stopping distance. HMI changes GREEN → AMBER → RED.

Scene 5 — Blind curve V2V

Truck B is hidden around a bend. Truck A receives its message before direct visual contact.

Scene 6 — Worker safety

Worker tag enters the simulated dynamic safety envelope. Truck displays WORKER DETECTED; tag gives vibration/buzzer.

Scene 7 — Communication failure

Disable cooperative communication. Display COOPERATIVE LINK LOST while local sensing remains active.

Network loss does not equal safety loss.

Scene 8 — Digital Twin

Dashboard displays truck locations, safety states, fog corridor, worker event, near-miss event, safety margin and event history.

39. THE JUDGE'S CORE QUESTION

“What is actually innovative here?”

Our innovation is not claiming that radar, V2V or thermal sensing are individually new. Existing research has demonstrated these technologies. Our innovation is the cost-optimized cooperative architecture that combines a local vehicle safety core with strategically placed Road Guardians, active worker tags and a physics-based dynamic safety bubble. This allows expensive sensing to be concentrated where it provides the most value, while keeping immediate safety decisions local to the vehicle.

40. “WHY CAN'T WE JUST STOP THE TRUCKS IN FOG?”

Stopping is the simplest safety response, but the problem statement also requires improved haulage efficiency and continuity. FogNet therefore does not assume that every low-visibility condition requires a complete shutdown. It measures the available safety margin, considers vehicle state and road conditions, and provides a conservative operating-speed recommendation. If the margin becomes unsafe, the system escalates the warning.

41. “WHY NOT JUST USE CAMERAS?”

Camera vision is useful but becomes less reliable as optical visibility deteriorates. FogNet therefore does not depend on a single optical sensor. Radar, motion sensing, cooperative information and roadside sensing provide complementary information. Thermal and optical sensing can be concentrated at strategic Road Guardian locations rather than duplicated on every vehicle.

42. “WHAT IF THE INTERNET FAILS?”

The Internet is not part of the immediate safety decision. The vehicle's local safety engine continues operating using its onboard sensors. Communication improves cooperative awareness, but losing communication does not disable the basic local safety layer.

43. “WHAT IS YOUR AI DOING?”

AI is used where prediction is useful—for example, fog trends, road risk patterns and near-miss analytics. Immediate collision-risk decisions are governed by deterministic safety calculations so that the driver can understand why a warning occurred.

44. “IS THIS AUTONOMOUS?”

No. The SIH prototype is a driver-assistance and cooperative safety system. It provides perception, calculation and warnings. Deeper vehicle-control integration is a future phase requiring OEM interfaces and appropriate validation.

Do not call the prototype fully autonomous.

45. “WHAT IS YOUR BIGGEST DIFFERENTIATOR?”

We are not trying to make every mining truck an expensive autonomous robot. We distribute intelligence across the mine: an affordable local safety core on each truck, higher-capability Road Guardians only at strategic locations, active worker tags, cooperative communication and a control-room risk layer. This creates a scalable safety network rather than an expensive sensor package on every vehicle.

46. SUCCESS CRITERIA FOR THE SIH PROTOTYPE

The prototype is successful if judges can physically observe:

- Fog condition changes
- Road Guardian detects the change
- Truck receives the warning
- Truck sensor detects an obstacle
- Safety margin changes with distance/speed
- Two trucks exchange information
- Worker tag generates a warning
- Communication can be intentionally disconnected
- Local safety still works

- Dashboard records the event

That is enough to establish the core FogNet concept.

47. FINAL SOLUTION STATEMENT

FogNet 4.1 transforms a conventional haul road into a cooperative safety environment.

Instead of asking: “Can the driver see through the fog?”

FogNet asks: “What information does the driver need when visibility is no longer sufficient?”

Vehicle perception detects immediate hazards. Road Guardians see strategically beyond individual vehicles. V2V/V2I shares information around blind sections. Worker tags make vulnerable ground personnel digitally visible. The Dynamic Safety Bubble converts speed, distance and road conditions into an understandable safety decision. The Digital Twin gives supervisors a real-time operational picture and learns from near-misses.

Local safety remains active even when the network fails.

FogNet does not attempt to eliminate fog. It eliminates the need for the mine to choose blindly between “STOP EVERYTHING” and “RUN BLIND.”

OPERATE WITH MEASURED, COOPERATIVE, CONDITION-AWARE SAFETY.

That is the purpose of FogNet.